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PHAM HOANG ANH

PREVENTIVE MEDICINE

HPV VACCINATION UPTAKE AND ASSOCIATED FACTORS AMONG VIETNAMESE WOMEN AGED 15–49 YEARS:

A SECONDARY DATA ANALYSIS OF THE 2020–2021 MICS 6 SURVEY

MEDICAL DOCTOR GRADUATION THESIS

COURSE 2020–2026

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1. Assoc. Prof. Pham Bich Diep **2. Dr. Pham Thanh Tung**

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Above all, I owe an immeasurable debt to my family for their unconditional love, patience, and sacrifice. You are my greatest motivation.

Hanoi, 2026

PHAM HOANG ANH

DECLARATION

I hereby solemnly declare that this thesis, entitled “HPV Vaccination Uptake and Associated Factors Among Vietnamese Women Aged 15–29 Years: A Secondary Data Analysis of the 2020–2021 MICS 6 Survey,” is my original work, conducted under the supervision of Assoc. Prof. Pham Bich Diep and Dr. Pham Thanh Tung. The data, statistical analyses, and findings presented herein are truthful and have not been previously submitted for any other degree or qualification at this or any other institution.

All referenced sources, including published literature, datasets, and methodological frameworks, are duly cited in accordance with the academic citation standards prescribed by Hanoi Medical University. The Vietnam MICS6 dataset used in this study was accessed in compliance with the official data use agreement and ethical protocols.

I take full responsibility for the integrity of the research presented in this thesis.

Hanoi, 2026

Student

PHAM HOANG ANH

TABLE OF CONTENTS

ABSTRACT

Background. Despite the proven efficacy of HPV vaccination, substantial disparities in awareness and uptake persist in low- and middle-income countries. In Vietnam, national evidence on the magnitude and determinants of these inequalities among young women remains limited.

Objectives. This study aimed to (1) estimate the weighted prevalence of HPV vaccine awareness and vaccination among Vietnamese women aged 15–29; (2) quantify socioeconomic inequalities using the concentration index; and (3) decompose the observed inequality via the Erreygers method.

Methods. A cross-sectional analysis of the Vietnam Multiple Indicator Cluster Survey 2020–2021 (MICS6) was conducted. The analytic sample comprised 4,102 women aged 15–29. Survey-weighted descriptive statistics, bivariate chi-squared tests, Poisson regression yielding prevalence ratios (PR), the Wagstaff concentration index, and Erreygers decomposition were employed.

Results. Overall, 62.4% of women had heard of HPV vaccination, while 7.5% had received the vaccine. The concentration indices for awareness (CI = 0.149, p ¡ 0.001) and vaccination (CI = 0.389, p ¡ 0.001) indicated significant pro-rich inequality. Wealth was the largest contributor, explaining 38.8% and 51.4% of the inequality in awareness and vaccination, respectively. Education, rural residence, and mass media exposure were substantial contributors.

Conclusion. Marked socioeconomic inequalities characterize HPV vaccine awareness and uptake among young Vietnamese women. Targeted interventions focusing on economically disadvantaged, less educated, rural, and ethnic minority women are urgently needed to advance health equity in cervical cancer prevention.

Keywords: HPV vaccination; socioeconomic inequality; concentration index; Erreygers decomposition; MICS; Vietnam.

LIST OF ABBREVIATIONS

Abbreviation	Full term
CI	Concentration Index
CIN	Cervical Intraepithelial Neoplasia
DHS	Demographic and Health Survey
ECI	Erreygers-Corrected Concentration Index
EPI	Expanded Programme on Immunization
GAVI	Global Alliance for Vaccines and Immunisation
GLM	Generalized Linear Model
HPV	Human Papillomavirus
ICT	Information and Communication Technology
IRB	Institutional Review Board
LMIC	Low- and Middle-Income Country
MICS	Multiple Indicator Cluster Survey
OR	Odds Ratio
PR	Prevalence Ratio
PSU	Primary Sampling Unit
SES	Socioeconomic Status
UNICEF	United Nations Children's Fund
VIA	Visual Inspection with Acetic Acid
WHO	World Health Organization

LIST OF TABLES

LIST OF FIGURES

INTRODUCTION

Cervical cancer is a defining metric of global health inequality. Despite being entirely preventable through prophylactic human papillomavirus (HPV) vaccination, it claimed 342,000 lives in 2020—nearly 90% of which occurred in low- and middle-income countries (LMICs) [1]. This disparity is not biological; it is fundamentally structural, rooted in the inequitable distribution of primary prevention [2].

In Vietnam, the national vaccination strategy remains heavily reliant on out-of-pocket payments in the private sector. Consequently, while the WHO mandates a 90% coverage target, actual uptake among Vietnamese women remains critically low [3,4]. More alarmingly, access to the vaccine—and even basic awareness of its existence—is strictly governed by socioeconomic fault lines: wealth, education, geography, and ethnicity [5, 6].

To dismantle these structural barriers, health policy must be guided by precise econometric quantification. The Erreygers decomposition framework provides a mathematically rigorous method to isolate and rank the exact determinants driving health disparities [7]. While this methodology is established globally, it has yet to be applied to national-level HPV vaccination data in Vietnam.

Leveraging the 2020–2021 Vietnam Multiple Indicator Cluster Survey (MICS6), this thesis focuses on 4,102 women aged 15–29 [8] to explicitly quantify the architecture of inequality in cervical cancer prevention among young Vietnamese women—the cohort most relevant to current HPV vaccination policy. By moving beyond simple descriptive statistics to robust econometric decomposition, this study aims to:

Research Objectives

1. Describe the weighted prevalence of HPV vaccine awareness and HPV vaccination among women aged 15–29 in Vietnam, stratified by sociodemographic

characteristics.

2. Quantify the magnitude of socioeconomic inequality in HPV vaccine awareness and vaccination using the concentration index.

3. Identify and quantify the contribution of socioeconomic determinants to the observed inequality through Erreygers decomposition analysis.

The findings are expected to provide actionable evidence for policymakers and program planners aiming to design equitable cervical cancer prevention strategies in Vietnam and comparable LMIC settings.

CHAPTER 1
LITERATURE REVIEW

1.1 Human papillomavirus and cervical cancer

1.1.1 Epidemiology of cervical cancer

Cervical cancer is the fourth most frequently diagnosed malignancy and the fourth leading cause of cancer death among women globally. According to GLOBOCAN 2020 estimates, approximately 604,000 new cases and 342,000 deaths were attributed to cervical cancer worldwide, with the vast majority of this burden concentrated in sub-Saharan Africa, South-East Asia, and Latin America [1]. In Vietnam, cervical cancer ranks among the top five cancers in women, with an estimated age-standardized incidence rate of 7.1 per 100,000 women-years and approximately 4,100 new cases diagnosed annually [9].

The striking geographic disparity in cervical cancer burden reflects fundamental inequities in access to primary prevention through HPV vaccination, secondary prevention through screening, and tertiary care through timely treatment of precancerous and cancerous lesions [2]. In high-income countries where organized screening programs have been established for decades, cervical cancer incidence has declined by 50–70%; by contrast, many LMICs continue to experience rising trends [10].

1.1.2 HPV virology and natural history

Human papillomavirus (HPV) is a small, non-enveloped, double-stranded DNA virus belonging to the *Papillomaviridae* family. Over 200 HPV genotypes have been identified, of which approximately 40 infect the anogenital mucosa. These are classified as either low-risk (e.g., HPV 6, 11) or high-risk (e.g., HPV 16, 18, 31, 33, 45, 52, 58) based on their oncogenic potential [11]. HPV types 16 and 18 alone account for approximately 70% of all cervical cancer cases globally [12].

The natural history of HPV-related cervical carcinogenesis follows a well-characterized multi-step pathway: initial HPV infection of the cervical transformation zone, viral persistence, progression through cervical intraepithelial neoplasia grades 1–3 (CIN1–CIN3), and ultimately invasive carcinoma. While most HPV infections are transient and resolve within 12–24 months through effective immune clearance, persistent infection with high-risk genotypes—particularly in the setting of immunosuppression or co-infections—can drive neoplastic transformation over a period of 10–20 years [13].

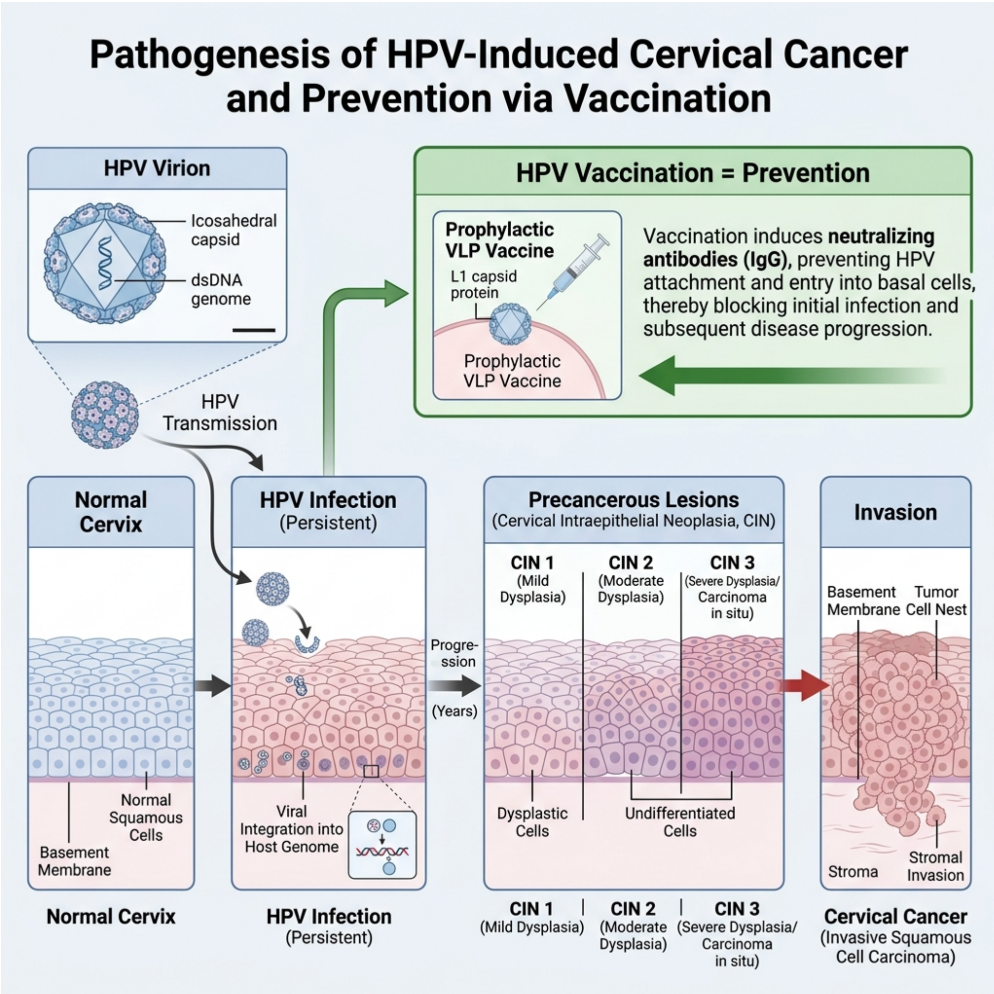

Figure 1.1. Natural history of HPV infection and cervical cancer development. HPV vaccination provides primary prevention by blocking initial infection, while screening programs enable secondary prevention through early detection and treatment of precancerous lesions.

1.1.3 HPV vaccines: types and efficacy

Three prophylactic HPV vaccines have been licensed and are currently available worldwide:

- **Bivalent vaccine (Cervarix®)**: targets HPV types 16 and 18.
- **Quadrivalent vaccine (Gardasil®)**: targets HPV types 6, 11, 16, and 18.
- **Nonavalent vaccine (Gardasil 9®)**: targets HPV types 6, 11, 16, 18, 31, 33, 45, 52, and 58, potentially preventing up to 90% of cervical cancers.

Large-scale randomized controlled trials and post-licensure surveillance data have demonstrated vaccine efficacy exceeding 90% against persistent infection and precancerous lesions caused by vaccine-targeted HPV types when administered prior to HPV exposure [14, 15]. The WHO currently recommends a one- or two-dose schedule for girls aged 9–14 years as the primary target population, with catch-up vaccination for females aged 15–26 [16].

Population-level impact has been substantial in countries with high vaccine coverage. A 2019 meta-analysis of data from 60 million individuals across 14 high-income countries demonstrated significant reductions in HPV 16/18 infections (83% among girls aged 13–19), anogenital warts (67%), and CIN2+ lesions (51%) within 5–8 years of vaccine introduction [15].

1.2 HPV vaccination globally and in Vietnam

1.2.1 Global vaccination landscape

As of 2023, over 140 countries had introduced HPV vaccination into their national immunization programs. However, global coverage remains uneven: while many high-income countries have achieved coverage rates exceeding 70%, coverage in LMICs—where the disease burden is greatest—remains below 15% [3]. The COVID-19 pandemic further disrupted vaccination programs, leading to an estimated

3.5 million missed doses worldwide between 2020 and 2022 [17].

The persistent gap between vaccine availability and actual uptake in LMICs reflects a complex interplay of supply-side factors (cost, cold-chain logistics, health system capacity) and demand-side barriers (awareness, cultural beliefs, misinformation, distance to health facilities) [18].

1.2.2 HPV vaccination in Vietnam

Vietnam's engagement with HPV vaccination began with a GAVI-supported demonstration project in 2006, initially targeting girls aged 11–12 years in selected provinces. The program was subsequently expanded through the national Expanded Programme on Immunization (EPI), although nationwide roll-out has proceeded gradually due to logistical and financial constraints [4].

As of 2023, HPV vaccination in Vietnam remains partially dependent on out-of-pocket payment in many provinces, which disproportionately disadvantages women from lower socioeconomic backgrounds. The estimated national coverage among age-eligible females remains below 30%, with marked urban–rural and inter-provincial disparities [5, 19]. These patterns underscore the need for equity-focused analyses to inform targeted intervention strategies.

1.3 Socioeconomic inequality in health: concepts and measurement

1.3.1 Defining health inequality

Health inequality refers to systematic, avoidable, and unjust differences in health outcomes or determinants across population groups defined by social, economic, demographic, or geographic characteristics [20]. When these differences arise from modifiable socioeconomic circumstances rather than biological variation, they are termed health *inequities* and carry a distinct normative imperative for policy action [21].

In the context of HPV vaccination, socioeconomic inequality manifests as differential awareness of and access to vaccination services across wealth strata. Such inequality is particularly concerning because it amplifies existing disparities: women who are poorest and least educated—and who often face the highest cervical cancer risk due to limited screening access—are also the least likely to be reached by vaccination programs [22].

1.3.2 The concentration index

The concentration index (CI) is the most widely used summary measure of socioeconomic inequality in health. Formally, the CI is defined as twice the area between the concentration curve and the 45-degree line of equality [23]:

$$CI = \frac{2}{\mu}\mathrm{cov}(y_i,\, r_i) \tag{1.1}$$

where y_i is the health variable for individual i, r_i is the fractional rank of individual i in the socioeconomic distribution, and μ is the population mean of y.

The CI ranges from -1 to $+1$. A positive value indicates that the health variable is concentrated among the wealthier (pro-rich inequality), a negative value indicates concentration among the poorer (pro-poor inequality), and zero implies perfect equality.

1.3.3 The Erreygers correction and decomposition

For bounded health variables (such as binary indicators of vaccination status), the standard CI may yield misleading comparisons across populations with different means. Erreygers (2009) proposed a corrected concentration index that satisfies mirror, level independence, and cardinal invariance properties [7]:

$$E(y) = \frac{4\mu}{b-a} \cdot CI \tag{1.2}$$

where a and b are the lower and upper bounds of the health variable (0 and 1 for binary outcomes).

The Erreygers-corrected CI can be decomposed into the contributions of individual determinants using the relationship:

$$E(y) = 4\sum_{k} \beta_k \bar{x}_k \cdot CI_k + \text{residual} \tag{1.3}$$

where β_k is the marginal effect of covariate x_k from a probit model, $\bar{x}_k$ is the covariate mean, and CI_k is the concentration index of the covariate with respect to socioeconomic rank. This decomposition enables researchers to quantify the percentage contribution of each factor—such as education, wealth, or geographic location—to the total observed inequality [24, 25].

1.4 Evidence on determinants of HPV awareness and vaccination

A growing body of literature has examined factors associated with HPV vaccine awareness and uptake in LMIC settings. Key findings from systematic reviews and country-specific studies include:

Socioeconomic status. Higher household wealth is consistently associated with greater awareness and higher vaccination rates. Studies in sub-Saharan Africa, South-East Asia, and Latin America have documented wealth-related gradients of 2–5 fold in vaccination coverage [22, 26].

Education. Women with secondary or higher education are significantly more likely to have heard of HPV vaccination and to have been vaccinated, reflecting both greater health literacy and enhanced access to information [4, 27].

Urban–rural residence. Urban women generally have higher awareness and vaccination rates, likely due to proximity to health facilities, greater media exposure, and higher density of vaccination service points [5, 18].

Ethnicity. In multi-ethnic societies, ethnic minority women often exhibit lower awareness and uptake, attributable to language barriers, cultural beliefs, geographic isolation, and systemic marginalization [6].

Mass media and ICT exposure. Exposure to newspapers, radio, television, internet, and mobile communication has been positively associated with health knowledge and preventive behavior uptake in numerous health domains, including vaccination [28, 29].

Age and marital status. Younger women and those who are unmarried may be more receptive to vaccination messages, though actual uptake patterns vary by country and program design [3].

1.5 Research gap and rationale

While the determinants of HPV vaccination have been extensively studied in high-income settings, several critical gaps remain in the Vietnamese context:

1. **Lack of inequality-focused analysis.** Most Vietnamese studies report prevalence and associations but do not quantify the magnitude of socioeconomic inequality using standardized indices.

2. **Limited decomposition evidence.** No nationally representative study has decomposed the sources of inequality in HPV awareness and vaccination in Vietnam using the Erreygers method.

3. **Awareness–vaccination gap.** The phenomenon whereby women are aware of the vaccine but fail to receive it—the "awareness–vaccination gap"—has received limited attention, yet it represents a distinct policy-relevant outcome.

4. **Outdated data.** Previous Vietnamese studies relied on smaller regional samples or earlier survey rounds. The MICS6 dataset (2020–2021) provides the most recent nationally representative evidence available.

1.6 Conceptual framework

Building on the social determinants of health framework (WHO Commission, 2008) and the behavioral model of health services utilization (Andersen, 1995), this study posits that HPV vaccine awareness and uptake are shaped by a constellation of predisposing, enabling, and reinforcing factors operating at the individual, household, and community levels.

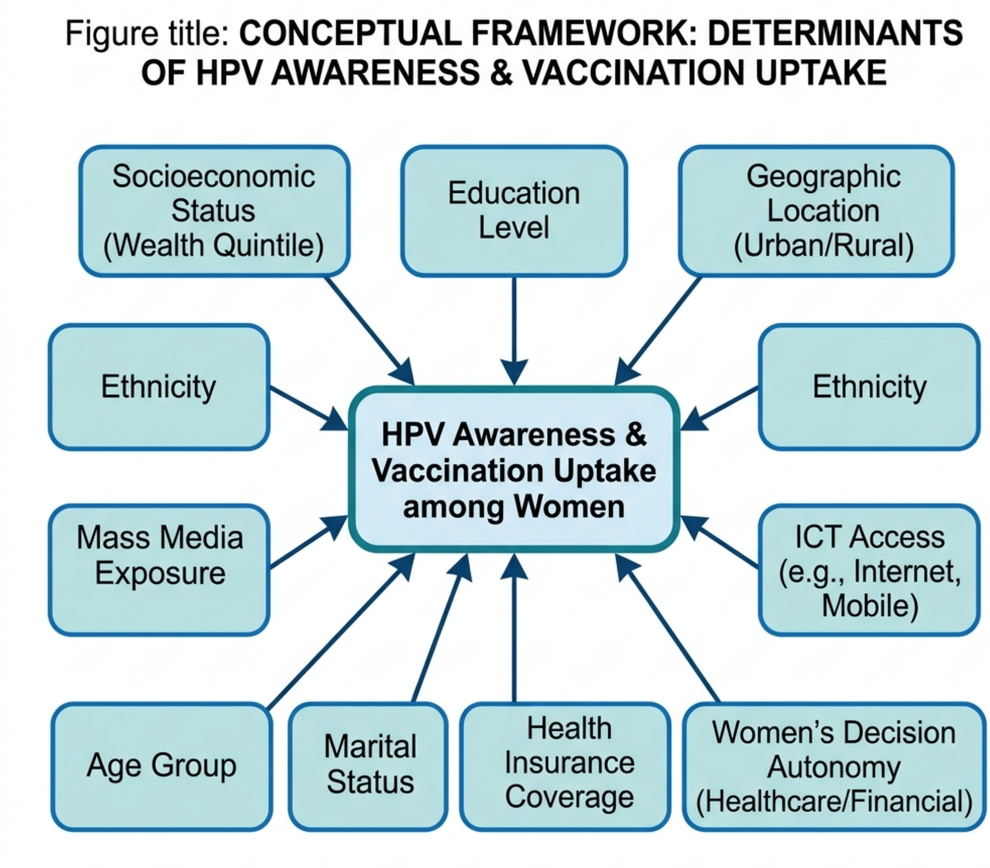

Figure 1.2. Conceptual framework illustrating the determinants of HPV vaccine awareness and vaccination among women aged 15–29 in Vietnam. Socioeconomic position, education, geographic location, ethnicity, and information exposure collectively shape both awareness and uptake, generating the observed inequality patterns.

As depicted in Figure 1.2, the framework identifies the following key pathways:

- **Structural determinants**: wealth quintile, education level, and urban/rural residence establish the fundamental socioeconomic gradient.
- **Intermediary determinants**: health insurance coverage, mass media and ICT

exposure, and decision-making autonomy mediate the relationship between structural position and health behavior.

- **Individual characteristics**: age, marital status, sexual experience, and ethnicity modify vulnerability and exposure patterns.

This framework guides the selection of covariates for the multivariable Poisson regression and provides the theoretical basis for interpreting the Erreygers decomposition results presented in Chapter 3.

Crucially, the framework conceptualizes the awareness–vaccination gap not merely as a sequential cognitive failure, but as a structural bottleneck where intermediary determinants exert differential pressures. While mass media and digital information exposure can successfully dismantle initial knowledge barriers, they are severely insufficient to overcome the profound financial and logistical hurdles of actual vaccine procurement in a predominantly privatized delivery system. This fundamental asymmetry dictates that the epidemiological predictors of awareness will diverge significantly from the predictors of final vaccine uptake.

Furthermore, the model acknowledges the intersectionality of these health determinants. For instance, ethnic minority status in Vietnam frequently intersects with geographic remoteness, language barriers, and lower baseline wealth, thereby compounding structural disadvantages at multiple levels of the health-seeking process. By mapping these interconnected pathways, the framework ensures that our subsequent econometric decomposition does not merely isolate abstract statistical associations. Instead, it precisely locates the actionable public health levers—whether universal health financing, targeted community outreach, or educational equity—required to dismantle the systemic inequities that sustain cervical cancer vulnerability among Vietnamese women.

CHAPTER 2

MATERIALS AND METHODS

2.1 Study design

This study employed a cross-sectional, secondary data analysis design, utilizing data from the Vietnam Multiple Indicator Cluster Survey (MICS6), conducted in 2020–2021 by the General Statistics Office of Vietnam in collaboration with UNICEF. MICS6 is a nationally representative household survey designed to generate disaggregated data on the situation of children and women, employing a two-stage stratified cluster sampling methodology [8].

The survey covered all 63 provinces and centrally-governed cities of Vietnam, with data collection conducted between January 2020 and June 2021. The Women's Questionnaire was administered through face-to-face interviews with all women aged 15–49 years residing in selected households, covering topics including reproductive health, child health, education, media exposure, and—critically for this study—knowledge of and experience with HPV vaccination.

2.2 Study participants

2.2.1 Inclusion criteria

- Women aged 15–29 years (corresponding to MICS age groups WAGE 1–3) who completed the Women's Individual Questionnaire.
- Valid sampling weights, primary sampling unit (PSU), and stratification information available for survey-weighted analysis.

2.2.2 Exclusion criteria

- Women with missing sampling weights (`wmweight`), PSU identifiers, or stratum indicators, as these preclude valid survey-weighted estimation.

- Records with indeterminate responses on both outcome variables (HPV awareness and vaccination status).

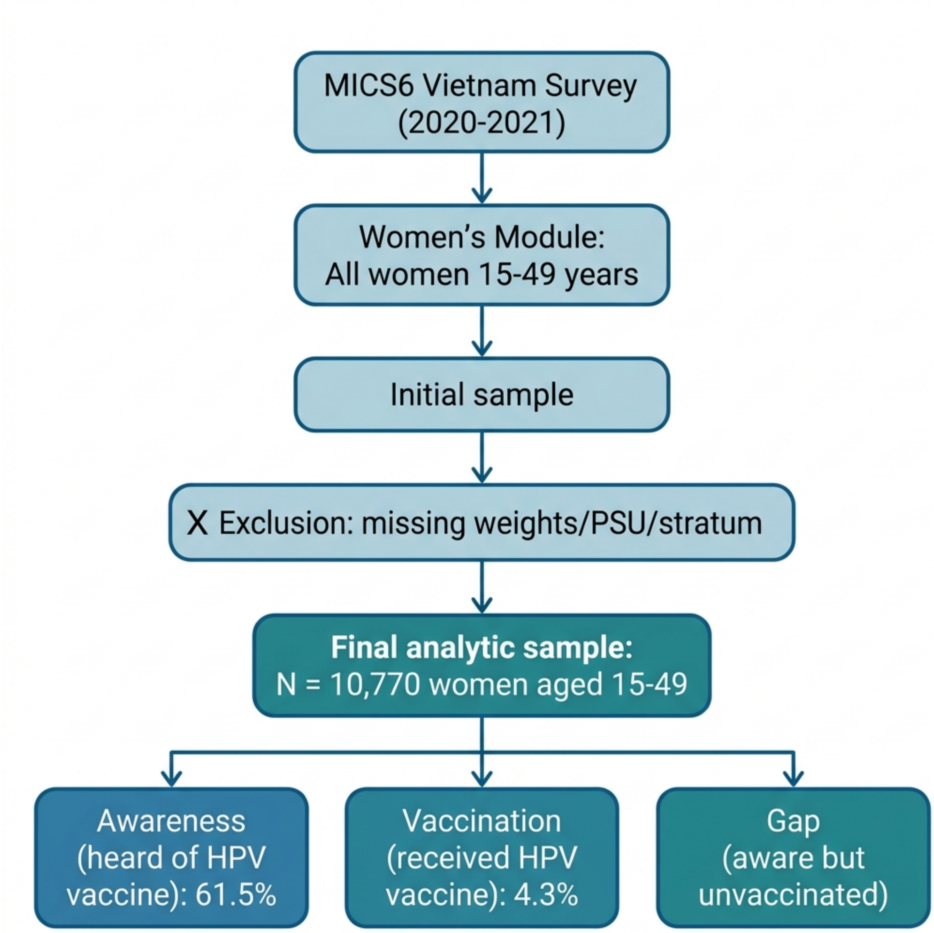

Figure 2.1. Study design and participant selection flowchart. From the MICS6 Women's Module, the final analytic sample comprised 4,102 women aged 15–29 after excluding records with missing survey design variables.

2.3 Sample size

The analytic sample consisted of all eligible women from the MICS6 survey who met the inclusion criteria. After applying exclusion criteria, the final sample comprised **N = 4,102** women aged 15–29 years. This sample size provides adequate statistical power for detecting inequality patterns and conducting multivariable regression and decomposition analyses. No additional sample size calculation was performed, as this study utilized the complete available dataset from a nationally representative survey.

2.4 Study variables

2.4.1 Outcome variables

Three binary outcome variables were constructed from the MICS6 cervical cancer prevention module:

1. **HPV vaccine awareness** (`ccp5_bin`): Derived from MICS item CCP5 ("Have you ever heard of HPV vaccination?"). Coded as 1 = Yes, 0 = No.

2. **HPV vaccination status** (`ccp6_bin`): Derived from MICS item CCP6 ("Have you ever received the HPV vaccine?"). Coded as 1 = Yes, 0 = No or Don't Know. Women who reported not having heard of HPV vaccination (CCP5 = No) were classified as unvaccinated (CCP6 = 0), consistent with the MICS skip pattern.

3. **Awareness–vaccination gap** (`gap`): Defined among aware women only (CCP5 = Yes). Coded as 1 if the woman was aware but unvaccinated, 0 if aware and vaccinated. This variable captures the "knowing–doing gap" relevant to behavioral intervention design.

2.4.2 Explanatory variables

Based on the conceptual framework (Section 1.6) and available MICS6 data, the following covariates were included:

2.5 Data analysis

All analyses were performed using Stata version 17 (StataCorp LLC, College Station, TX) with the `svy:` prefix to account for the complex survey design (stratification, clustering, and sampling weights). The analytical strategy followed a sequential five-step approach:

Table 2.1. Summary of explanatory variables used in the analysis

Variable	Categories	Operational Definition
Age group (WAGE)	15–19, 20–24, 25–29	MICS standard 5-year age groups
Residence (area)	Urban, Rural	MICS household classification
Ethnicity (minority)	Kinh/Hoa (No), Ethnic minority (Yes)	MICS variable ethnicity2
Marital status (MSTATUS)	Currently married/union, Formerly married, Never married	MICS marital status variable
Education (edu)	Primary or less, Lower secondary, Upper secondary+	Collapsed from MICS welevel
Wealth quintile (ses5)	Q1 (poorest) to Q5 (richest)	MICS wealth index quintiles
Health insurance (insu)	Insured (No), Uninsured (Yes)	MICS insurance variable
Mass media exposure (mme)	No, Yes	Any exposure to newspaper, radio, or TV
ICT exposure (ict)	No, Yes	Any use of computer, internet, or mobile
Decision autonomy (dvindex)	Low (No), High (Yes)	All 5 DV1 items answered "jointly"
Sexual experience (sb1n)	No, Yes	Ever had sexual intercourse (SB1)

2.5.1 Step 1: Descriptive statistics

Survey-weighted proportions with 95% confidence intervals were computed for all outcome and explanatory variables to characterize the study population and estimate the prevalence of HPV awareness and vaccination.

2.5.2 Step 2: Bivariate analysis

The association between each explanatory variable and the outcome variables was assessed using survey-weighted Pearson chi-squared tests with Rao–Scott corrections for complex survey design. Row percentages with 95% confidence intervals were reported. Statistical significance was set at $\alpha = 0.05$.

2.5.3 Step 3: Concentration index

The Wagstaff concentration index (CI) provides a rigorous econometric quantification of socioeconomic inequality in health outcomes [23]. In this study, the

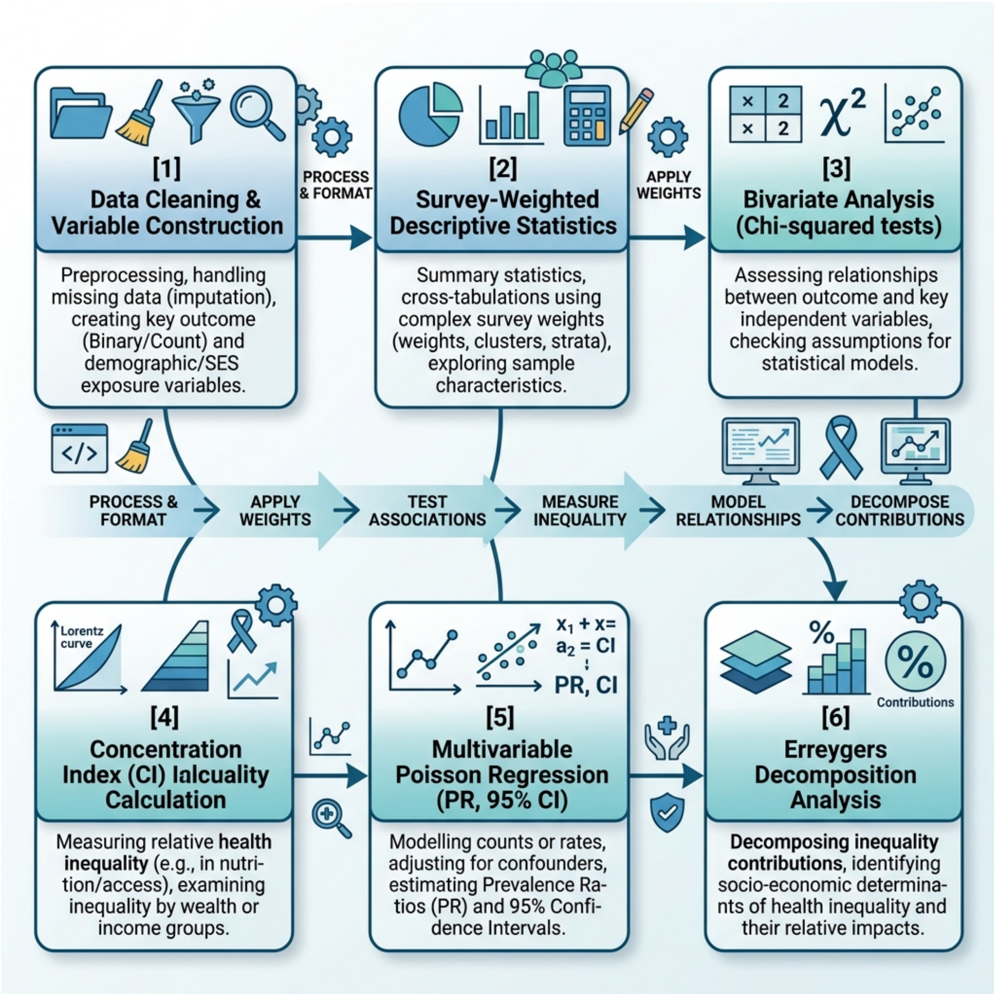

Figure 2.2. Statistical analysis pipeline. The six-stage analytical approach progresses from data preparation through descriptive analysis, bivariate testing, inequality quantification, multivariable modeling, and decomposition.

standard CI was formulated mathematically as both the scaled covariance between the health outcome and the fractional rank in the living standards distribution, and geometrically as the area integral:

$$CI = \frac{2}{\mu}\mathrm{cov}(y_i, r_i) = \frac{2}{\mu}\int_0^1 \big(y(r) - \mu\big) r \, dr \tag{2.1}$$

where y_i represents the binary outcome for individual i, r_i is the continuous fractional rank in the socioeconomic distribution (household wealth score), and μ is the survey-weighted population mean of the outcome. Geometrically, the CI equals twice the area between the concentration curve and the line of equality. A positive CI ($0 <$

$CI \leq 1$) indicates pro-rich inequality, whereas a negative CI ($-1 \leq CI < 0$) signifies pro-poor concentration.

2.5.4 Step 4: Multivariable Poisson regression

Because the prevalence of HPV awareness exceeded the 10% threshold, logistic regression would mathematically yield odds ratios that severely overestimate the true relative risk [30]. Consequently, we employed modified Poisson regression—a generalized linear model (GLM) featuring a logarithmic link function and a Poisson distribution—to directly estimate adjusted prevalence ratios (PR). To correct for the violation of the Poisson assumption (variance equal to the mean) inherent in binary data, robust Huber-White sandwich variance estimators were applied. The formal model specification is:

$$\log\big[E(Y_i \mid \mathbf{X}_i)\big] = \mathbf{X}_i^\top \boldsymbol{\beta} \quad \text{with robust variance:} \quad \widehat{\mathbf{V}}(\hat{\boldsymbol{\beta}}) = \mathbf{D}^{-1}\mathbf{M}\mathbf{D}^{-1} \tag{2.2}$$

where $E(Y_i \mid \mathbf{X}_i)$ is the expected probability of the outcome given the covariate vector $\mathbf{X}_i$, and $\boldsymbol{\beta}$ is the vector of coefficients. The exponentiated coefficients ($\exp(\beta_k)$) provide the direct prevalence ratios. The matrix D represents the expected information matrix, and M is the outer product of the empirical gradients. The reference categories were: age 25–29, currently married, upper secondary+ education, Q5 (richest), and the absence category for binary covariates.

A sensitivity analysis using logistic regression (yielding odds ratios) was conducted to assess the robustness of findings.

2.5.5 Step 5: Erreygers decomposition

A fundamental mathematical limitation of the standard Wagstaff CI is that its absolute bounds contract as the mean of the binary outcome deviates from 0.5, distorting comparative analyses across populations. To circumvent this scale dependence, we utilized the Erreygers Concentration Index (ECI), which restores the normative limits to $[-1, 1]$ via the transformation:

$$ECI(y) = 4\mu(1-\mu)CI(y) \tag{2.3}$$

The ECI was subsequently decomposed to isolate the specific econometric contributions of each explanatory variable [7]. Given a generalized linear mapping via a probit link, the exact contribution of each covariate k is calculated as the product of its partial marginal effect, its weighted mean, and its own concentration index:

$$\text{Contribution}_k = 4 \cdot \left(\frac{\partial E(y \mid \mathbf{x})}{\partial x_k} \right) \cdot \bar{x}_k \cdot CI_{x_k} = 4 \cdot \beta_k^{AME} \cdot \bar{x}_k \cdot CI_{x_k} \tag{2.4}$$

where β_k^{AME} is the average marginal effect from a probit model, $\bar{x}_k$ is the weighted mean of the covariate, and CI_{x_k} is the Erreygers-corrected concentration index of the covariate with respect to the wealth ranking. The percentage contribution of each covariate was computed as:

$$\text{Percentage}_k = \frac{\text{Contribution}_k}{ECI_{\text{total}}} \times 100 \tag{2.5}$$

Decomposition was performed separately for each of the three outcome variables: awareness, vaccination, and the awareness–vaccination gap.

2.5.6 Step 6: Stratified and sensitivity analyses

Additional analyses included:

- **Stratified regression** by urban/rural residence to examine whether determinants differed between settings.
- **Crude (unadjusted) Poisson regression** for each variable individually to compare crude and adjusted prevalence ratios.
- **Logistic regression sensitivity analysis** to verify that key findings were robust

to the choice of link function.

- **Gap model**: Poisson regression among aware women to identify factors associated with failure to translate awareness into vaccination action.

2.6 Ethical considerations

The MICS6 survey protocol was approved by the Institutional Review Board of the General Statistics Office of Vietnam and received ethical clearance from UNICEF's Division of Data, Analytics, Planning, and Monitoring. Written informed consent was obtained from all individual respondents prior to interview participation. For respondents under the age of 18, parental/guardian consent was also obtained.

This study constitutes a secondary analysis of de-identified, publicly available survey data. No direct contact with human subjects occurred. All data were accessed and analyzed in compliance with the MICS data use agreement.

CHAPTER 3

RESULTS

3.1 Characteristics of the study population

The final analytic sample comprised 4,102 women aged 15–29 years from the Vietnam MICS6 survey (2020–2021). Table 3.1 presents the survey-weighted distribution of sociodemographic characteristics.

The majority of women resided in rural areas (62.6%), belonged to the Kinh/Hoa ethnic group (85.0%), and half were never married (50.3%). Most had attained upper secondary education or higher (72.7%). Health insurance coverage was high (86.3%), as was mass media exposure (90.9%) and ICT access (97.9%).

Regarding the primary outcomes, 62.4% (95% CI: 59.8–64.9%) of women reported having heard of HPV vaccination, while only 7.5% (95% CI: 6.3–8.8%) had received the vaccine—revealing a substantial awareness–vaccination gap.

3.2 Bivariate analysis of HPV awareness

Table 3.2 presents the weighted prevalence of HPV vaccine awareness by sociodemographic characteristics.

A clear socioeconomic gradient emerged: awareness increased from 33.3% in the poorest quintile to 83.6% in the richest ($p < 0.001$). Awareness rose sharply with education (22.9% primary vs. 69.2% upper secondary+). Furthermore, awareness was substantially higher among urban (73.1%) compared to rural (56.0%) women. Ethnic minority women had awareness of just 36.1% compared to 67.0% among Kinh/Hoa women. Notably, health insurance was not significantly associated with awareness ($p = 0.233$).

Table 3.1. Weighted sociodemographic characteristics of the study population (N = 4,102)

Characteristic	**Category**	**%**
Age group	15–19	30.4
	20–24	29.7
	25–29	39.9
Residence	Urban	37.4
	Rural	62.6
Ethnicity	Kinh/Hoa	85.0
	Ethnic minority	15.0
Marital status	Currently married/union	46.4
	Formerly married/union	3.3
	Never married/union	50.3
Education	Primary or less	5.2
	Lower secondary	22.1
	Upper secondary+	72.7
Wealth quintile	Q1 (poorest)	19.2
	Q2	21.2
	Q3	21.9
	Q4	20.1
	Q5 (richest)	17.6
Health insurance	Insured	86.3
	Uninsured	13.7
Mass media exposure	Yes	90.9
	No	9.1
ICT exposure	Yes	97.9
	No	2.1
Decision autonomy	High	90.2
	Low	9.8
Ever had sexual intercourse	Yes	53.1
	No	46.9
Outcome variables		
HPV vaccine awareness	Yes	62.4
HPV vaccination	Yes	7.5

3.3 Bivariate analysis of HPV vaccination

Table 3.3 summarizes vaccination prevalence by sociodemographic factors.

The wealth gradient in vaccination was extremely steep: only 0.7% of women in the poorest quintile had been vaccinated, compared to 16.6% in the richest—a 24-fold difference. This transition from a 2.5-fold disparity in awareness to a 24-fold disparity in vaccination demonstrates that out-of-pocket costs and health system access are the true rate-limiting steps. Vaccination was highest among women aged 20–24 (10.0%),

Table 3.2. HPV vaccine awareness by sociodemographic characteristics (N = 4,102)

Characteristic	Aware (%)	95% CI	p-value
Age group			
15–19	48.3	44.2–52.4	<0.001
20–24	66.2	61.9–70.1	
25–29	70.3	67.0–73.4	
Residence			
Urban	73.1	69.0–76.8	<0.001
Rural	56.0	52.6–59.3	
Ethnicity			
Kinh/Hoa	67.0	64.2–69.8	<0.001
Ethnic minority	36.1	31.7–40.6	
Education			
Primary or less	22.9	16.0–31.6	<0.001
Lower secondary	49.2	44.3–54.1	
Upper secondary+	69.2	66.4–71.9	
Wealth quintile			
Q1 (poorest)	33.3	29.0–37.9	<0.001
Q2	58.5	53.6–63.3	
Q3	64.4	59.3–69.1	
Q4	73.5	68.8–77.8	
Q5 (richest)	83.6	79.5–87.0	
Mass media exposure			
No	39.1	32.8–45.7	<0.001
Yes	64.7	62.1–67.3	
Health insurance			
Insured	63.0	60.3–65.6	0.233
Uninsured	58.7	51.7–65.4	
Decision autonomy			
High	65.2	62.4–67.8	<0.001
Low	45.2	38.5–52.0	

likely reflecting opportunistic catch-up vaccination.

Figure 3.3 graphically illustrates this geographic divide. Urban residence significantly facilitates both awareness and actual vaccine receipt.

As depicted in Figure 3.4, educational attainment operates as a powerful structural determinant. Women with upper secondary education or higher demonstrate substantially greater vaccine uptake compared to those with primary education or less.

Figure 3.5 highlights the marginalization of ethnic minority women, with aware-

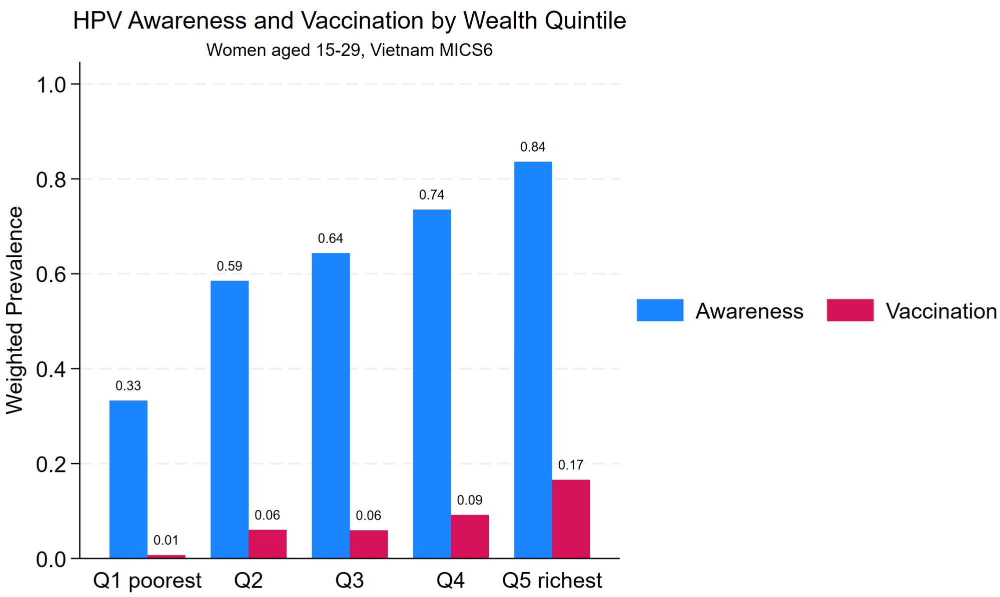

Figure 3.1. HPV vaccine awareness and vaccination prevalence by wealth quintile among women aged 15–29, Vietnam MICS6 (2020–2021). A steep socioeconomic gradient is evident for both outcomes.

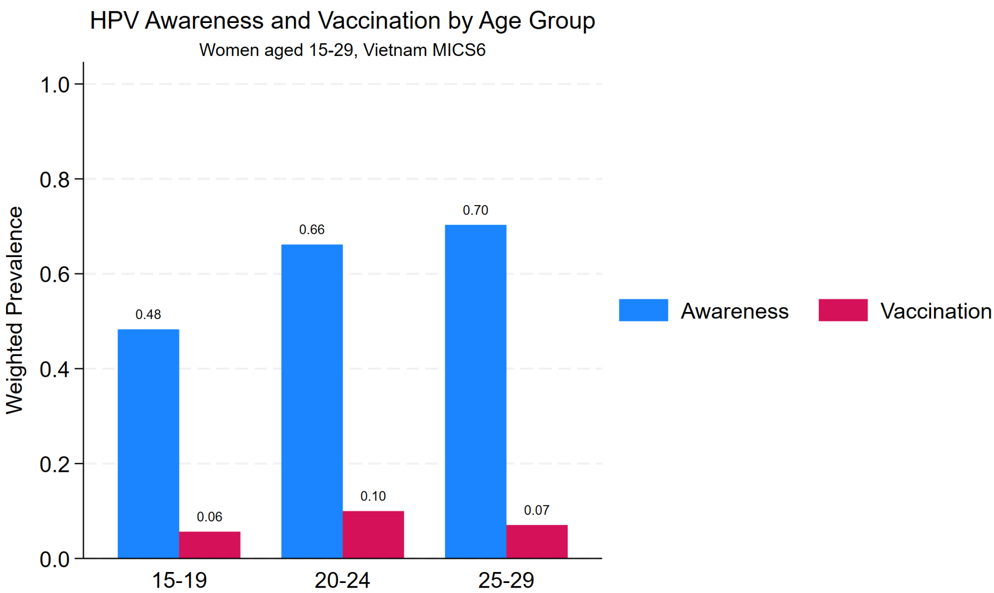

Figure 3.2. HPV vaccine awareness and vaccination by age group. Awareness peaks at age 25–29, while vaccination is highest among women aged 20–24.

ness at 36.1% and vaccination at just 1.6%.

Table 3.3. HPV vaccination status by sociodemographic characteristics (N = 4,102)

Characteristic	Vaccinated (%)	95% CI	p-value
Age group			
15–19	5.6	3.9–8.0	0.015
20–24	10.0	7.7–12.8	
25–29	7.0	5.5–8.9	
Residence			
Urban	11.4	9.1–14.1	<0.001
Rural	5.1	3.9–6.7	
Ethnicity			
Kinh/Hoa	8.5	7.2–10.1	<0.001
Ethnic minority	1.6	0.8–3.4	
Education			
Primary or less	0.8	0.1–4.0	<0.001
Lower secondary	2.4	1.3–4.3	
Upper secondary+	9.5	8.0–11.3	
Wealth quintile			
Q1 (poorest)	0.7	0.3–1.7	<0.001
Q2	6.0	4.1–8.9	
Q3	5.9	4.1–8.5	
Q4	9.2	6.7–12.4	
Q5 (richest)	16.6	12.7–21.3	

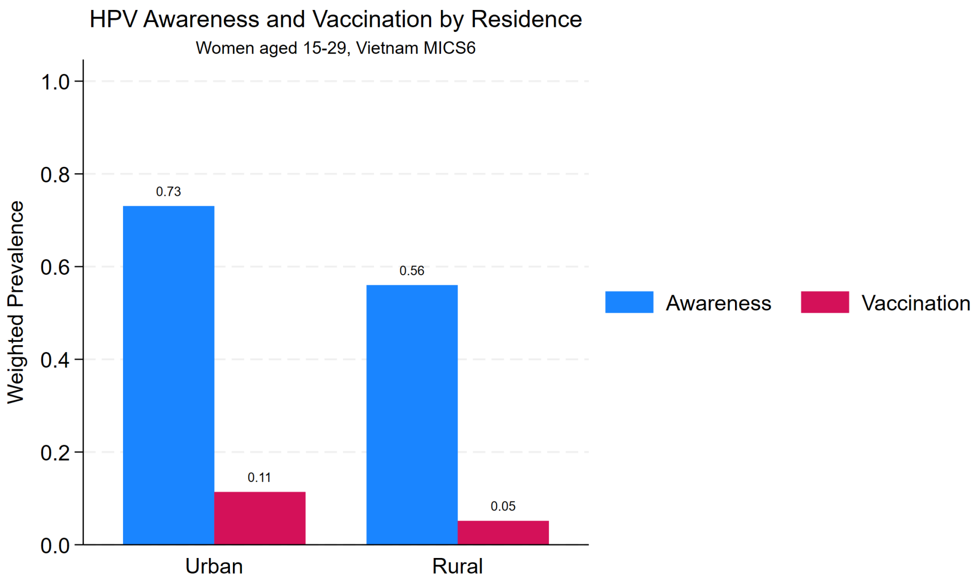

Figure 3.3. HPV awareness and vaccination by urban/rural residence. Urban women show substantially higher rates for both outcomes.

3.4 Socioeconomic inequality: concentration index

3.4.1 Overall concentration indices

Table 3.4 presents the overall concentration indices for each outcome.

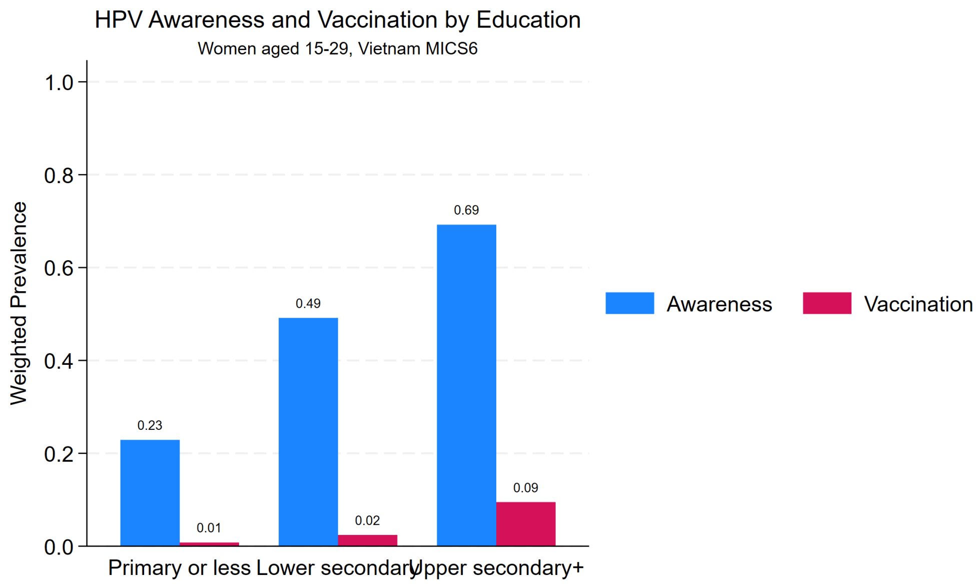

Figure 3.4. HPV awareness and vaccination by education level. A steep positive gradient is evident, with upper secondary+ women showing the highest rates.

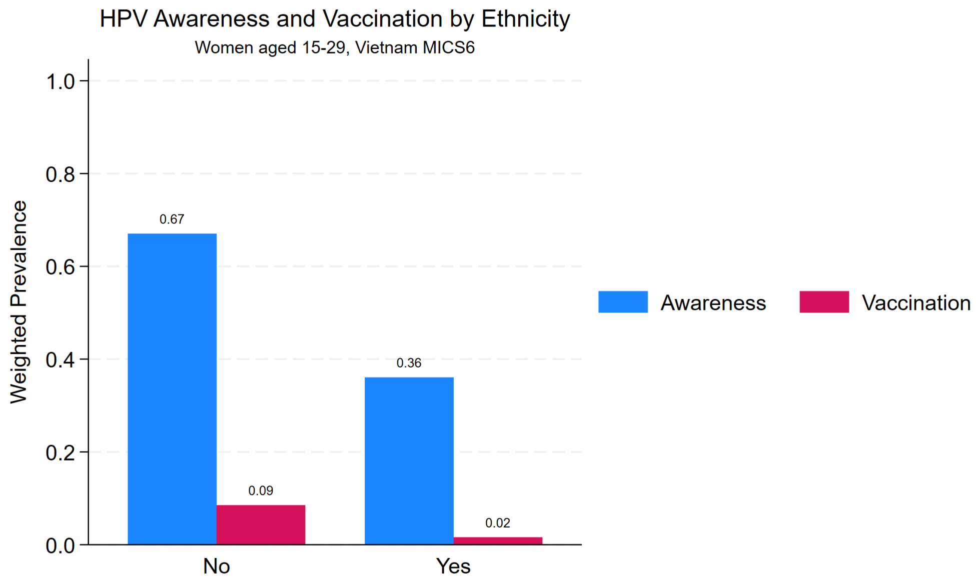

Figure 3.5. HPV awareness and vaccination by ethnicity. Ethnic minority women have substantially lower awareness and near-zero vaccination coverage.

All three concentration indices were statistically significant ($p < 0.001$). The positive CIs for awareness (0.149) and vaccination (0.389) indicate **pro-rich inequality**—both outcomes are concentrated among wealthier women. The magnitude of inequality was markedly greater for vaccination than for awareness, suggesting that

Table 3.4. Wagstaff concentration indices for HPV awareness, vaccination, and awareness–vaccination gap

Outcome	N	CI	SE	95% CI
Awareness	4,102	0.149	0.009	0.131–0.166
Vaccination	4,102	0.389	0.047	0.296–0.482
Gap	2,133	−0.035	0.006	−0.047–−0.023

CI = Wagstaff concentration index, ranked by household wealth score. Survey-weighted estimation using `conindex` in Stata. Positive values indicate pro-rich concentration; negative values indicate pro-poor concentration.

wealth-related barriers operate more strongly at the point of vaccine uptake than at the point of information exposure.

The negative CI for the awareness–vaccination gap (–0.035) indicates that the gap is concentrated among poorer women who, despite being aware, face additional barriers to actually receiving the vaccine.

3.4.2 Concentration curves

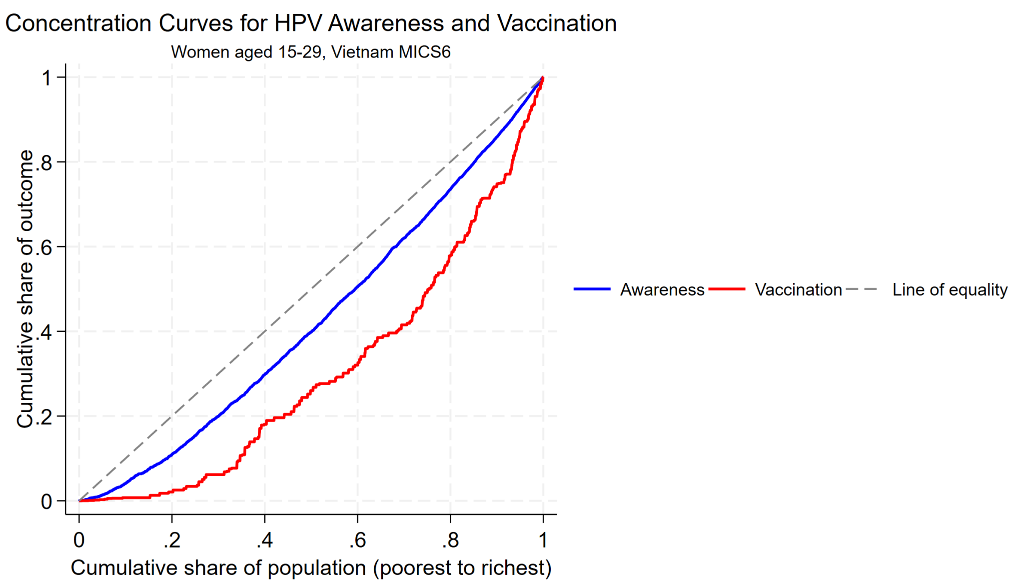

Figure 3.6. Concentration curves for HPV awareness and vaccination, ranked by household wealth score. Both curves lie below the line of equality, confirming pro-rich concentration. The vaccination curve departs further from equality, reflecting greater socioeconomic inequality in vaccine uptake.

As shown in Figure 3.6, both concentration curves lie below the diagonal line of equality, visually confirming that awareness and vaccination are disproportionately concentrated among wealthier women. The vaccination curve lies further from the equality line than the awareness curve, consistent with the larger CI value for vaccination (0.389 vs. 0.149).

3.5 Multivariable Poisson regression

Table 3.5 presents the adjusted prevalence ratios from multivariable survey-weighted Poisson regression for awareness and vaccination outcomes.

Table 3.5. Adjusted prevalence ratios (PR) from multivariable Poisson regression (N = 3,899)

	Awareness		Vaccination	
Variable	**PR**	**95% CI**	**PR**	**95% CI**
Rural residence (ref: Urban)				
Yes	0.933	0.867–1.005	0.780	0.547–1.112
Age group (ref: 25–29)				
15–19	0.626***	0.561–0.699	0.514*	0.305–0.865
20–24	0.918*	0.855–0.986	1.150	0.796–1.660
Education (ref: Upper secondary+)				
Primary or less	0.475***	0.330–0.683	0.327	0.060–1.770
Lower secondary	0.836***	0.754–0.927	0.422**	0.227–0.785
Wealth quintile (ref: Q5 richest)				
Q1 (poorest)	0.576***	0.480–0.691	0.118***	0.039–0.358
Q2	0.802***	0.726–0.885	0.585*	0.365–0.936
Q3	0.812***	0.743–0.888	0.437***	0.278–0.688
Q4	0.890**	0.829–0.956	0.631*	0.421–0.945
Mass media exposure (ref: No)				
Yes	1.147	0.981–1.340	1.960	0.763–5.034
Ethnic minority (ref: No)				
Yes	0.860*	0.745–0.992	0.743	0.307–1.801
Decision autonomy (ref: Low)				
High	1.234**	1.066–1.429	1.614	0.861–3.025

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$. PR = prevalence ratio. N = 3,899 after listwise deletion.

Awareness. After adjusting for all covariates, women in the poorest quintile had 42% lower awareness than the richest (PR = 0.576, 95% CI: 0.480–0.691). Primary education was associated with a 53% reduction (PR = 0.475, 95% CI: 0.330–0.683). Ethnic minority status (PR = 0.860) and younger age (15–19: PR = 0.626) were independently associated with lower awareness. High decision autonomy (PR = 1.234) was protective.

Vaccination. The socioeconomic gradient was substantially steeper. Women in the poorest quintile had an 88% lower prevalence of vaccination than the richest (PR = 0.118, 95% CI: 0.039–0.358). Women aged 15–19 had a 49% lower prevalence than those aged 25–29 (PR = 0.514). Lower secondary education showed a pronounced effect (PR = 0.422).

3.6 Erreygers decomposition analysis

3.6.1 Decomposition of awareness inequality

Table 3.6 presents the Erreygers decomposition results for HPV awareness inequality.

Table 3.6. Erreygers decomposition of socioeconomic inequality in HPV awareness (Erreygers CI = 0.371)

Variable	dy/dx	CI_x	Contribution	% of total
Q1 poorest	−0.302	−0.621	0.1440	**38.8**
Q2	−0.187	−0.342	0.0544	**14.6**
Rural	−0.050	−0.412	0.0514	**13.8**
Mass media	0.063	0.180	0.0413	**11.1**
Decision autonomy	0.102	0.089	0.0327	8.8
Never married	0.062	0.237	0.0296	8.0
Lower secondary	−0.102	−0.317	0.0285	7.7
Ethnic minority	−0.065	−0.413	0.0161	4.3

The Erreygers-corrected CI for awareness was 0.371, confirming pro-rich concentration. Household wealth (Q1 poorest) was the dominant structural determinant, generating **38.8%** of the total inequality. Geographic marginalization (rural residence) drove 13.8%, lack of mass media access drove 11.1%, and educational deficits contributed 7.7% of the disparity.

3.6.2 Decomposition of vaccination inequality

The poorest wealth quintile alone is responsible for generating **51.4%** of the total inequality in vaccine uptake. This confirms that financial insolvency overrides almost all other considerations at the point of care. Mass media exclusion generated 23.8% of the disparity, while rural isolation (17.1%) and educational deficits (12.3%)

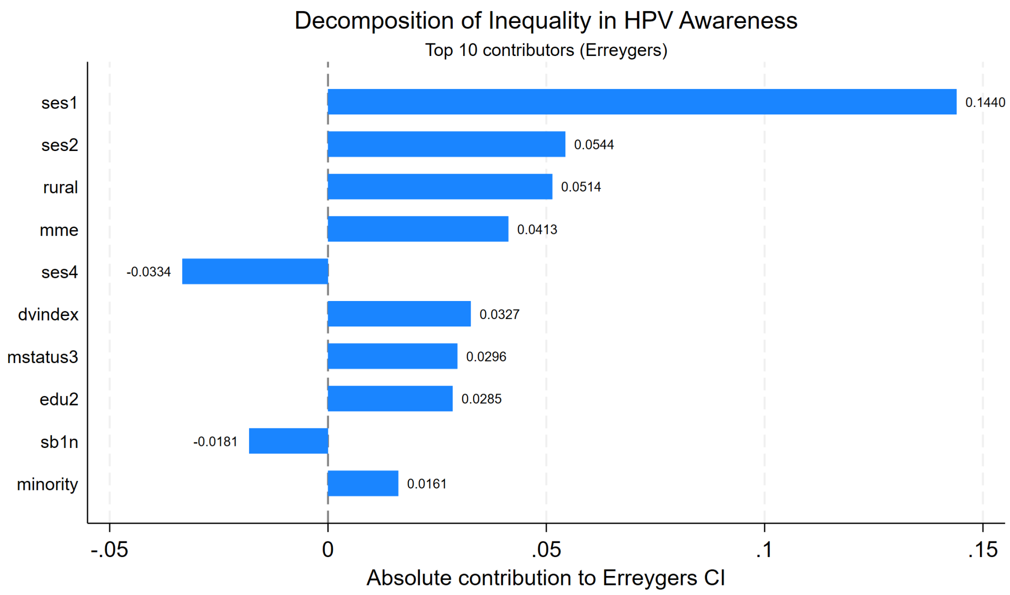

Figure 3.7. Top 10 contributors to socioeconomic inequality in HPV awareness (Erreygers decomposition). Wealth status, rural residence, mass media exposure, and education are the dominant drivers.

Table 3.7. Erreygers decomposition of socioeconomic inequality in HPV vaccination (Erreygers CI = 0.116)

Variable	dy/dx	CI_x	Contribution	% of total
Q1 poorest	−0.125	−0.621	0.0598	**51.4**
Mass media	0.042	0.180	0.0276	**23.8**
Rural	−0.019	−0.412	0.0199	**17.1**
Lower secondary	−0.051	−0.317	0.0143	**12.3**
Q2	−0.045	−0.342	0.0129	11.1
Sexual experience	−0.025	−0.223	0.0119	10.2
Decision autonomy	0.033	0.089	0.0105	9.0

acted as secondary structural barriers.

3.6.3 Decomposition of awareness–vaccination gap inequality

The Erreygers CI for the gap was −0.123, indicating pro-poor concentration: among aware women, those from lower socioeconomic strata were more likely to remain unvaccinated. The largest contributors were Q1 poorest (57.6%), mass media (35.1%), rural residence (16.4%), and lower secondary education (14.4%).

Figure 3.9 visualizes this critical failure point: among women who already possess adequate knowledge of the HPV vaccine, those in the poorest quintile face

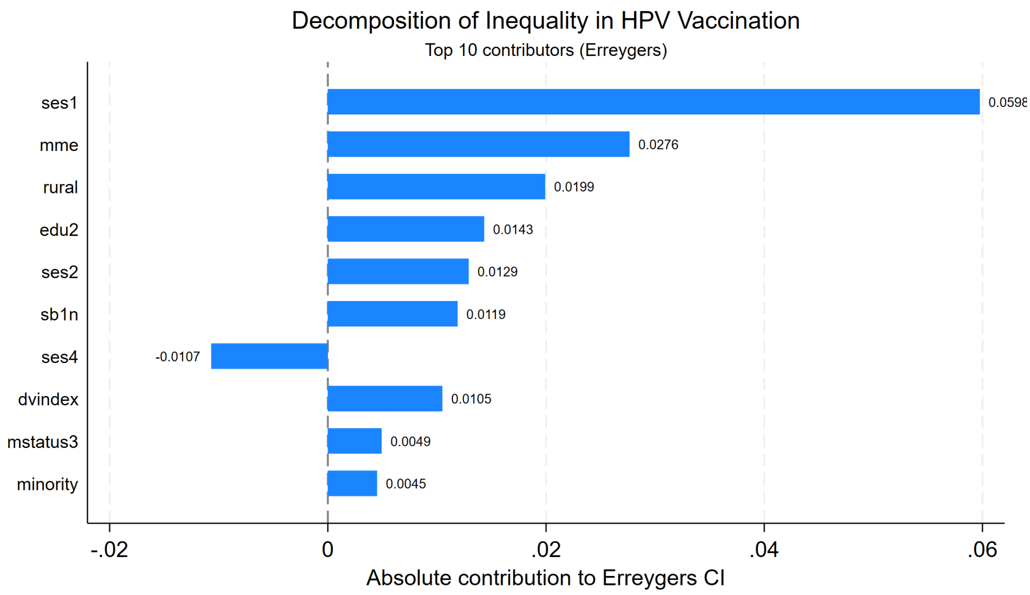

Figure 3.8. Top 10 contributors to socioeconomic inequality in HPV vaccination (Erreygers decomposition). Wealth status alone explains over half of the observed inequality.

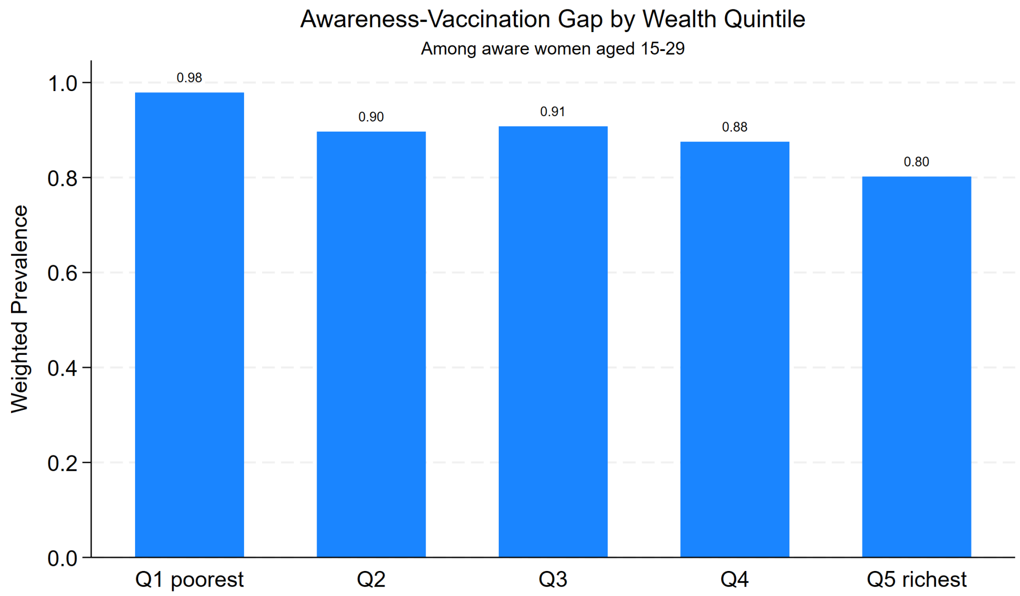

Figure 3.9. Awareness–vaccination gap by wealth quintile among aware women. The gap is widest among the poorest, reflecting barriers beyond awareness.

the widest gap between intent and action. This confirms that informational campaigns alone are insufficient; without concurrent financial subsidies or inclusion in the national Expanded Programme on Immunization (EPI), the poorest women remain structurally locked out of cervical cancer prevention.

CHAPTER 4
DISCUSSION

This study provides the first nationally representative quantification of socioeconomic inequality in HPV vaccine awareness and uptake among young women aged 15–29 in Vietnam. By integrating survey-weighted modeling with the Erreygers decomposition framework, we elucidate the structural determinants that drive the profound equity gap in cervical cancer prevention.

The knowing–doing chasm. Our analysis reveals a striking dichotomy between cognition and action: while 62.4% of women have heard of the HPV vaccine, a mere 7.5% have received it. This 55-percentage-point awareness–uptake gap underscores a fundamental failure of translation from health literacy to health behavior. Similar implementation bottlenecks have been documented across Southeast Asia—Thailand reports 58.3% awareness against sub-5% uptake [31], and the Philippines exhibits single-digit coverage despite 55.7% awareness [32]. In stark contrast, high-income nations with publicly financed, school-based programs routinely achieve coverage exceeding 70% [33]. The Vietnamese gap is therefore not merely informational but fundamentally structural, rooted in the absence of universal public financing and the prohibitive out-of-pocket costs of the vaccine series.

The anatomy of inequality. The concentration indices demonstrate severe pro-rich inequality, with the magnitude of disparity for vaccination (CI = 0.389) more than double that for awareness (CI = 0.149). This differential implies that while information diffuses across socioeconomic strata—albeit imperfectly—the actual receipt of the vaccine is profoundly constrained by barriers at the point of service delivery. This aligns with Penchansky and Thomas's access framework [34]. Information can diffuse relatively inexpensively through digital networks; receiving a vaccine, however, demands physical proximity to a health facility, the financial capacity to absorb high costs, and the operational autonomy to seek care—dimensions sharply patterned by socioeconomic position.

Critically, the negative CI for the awareness–vaccination gap (-0.035) reinforces this paradigm: among women already possessing knowledge of the vaccine, the poorest are disproportionately paralyzed from acting upon it. Policy interventions predicated solely on raising awareness will therefore inevitably fail to close the equity divide.

Wealth as the dominant engine of disparity. The Erreygers decomposition unequivocally isolates household wealth as the primary catalyst of inequity, explaining 38.8% of awareness inequality and an overwhelming 51.4% of vaccination inequality. This dominance mirrors global multi-country analyses of childhood immunization [25]. In the Vietnamese context, where the HPV vaccine series costs approximately 1.5–3.0 million VND in the private sector [4], the intervention effectively functions as a luxury good rather than a basic public health right. Adjusted multivariable models confirm this severe penalty: women in the poorest quintile face an 88% reduction in the probability of vaccination (PR = 0.118) compared to their wealthiest counterparts.

Education: the second pillar of inequality. Education emerged as an important driver, contributing 7.7% and 12.3% to awareness and vaccination inequality, respectively. The dose-response gradient is steep: primary-educated women exhibit a 53% reduction in awareness likelihood (PR = 0.475) and lower secondary women show a 58% reduction in vaccination (PR = 0.422) relative to those with upper-secondary education or higher. These associations likely operate through enhanced health literacy and stronger integration into social networks that normalize preventive health behaviors [35].

The role of the information ecosystem and geographic fault lines. Mass media exposure contributed 11.1% and 23.8% to the inequality in awareness and vaccination, respectively. Women lacking exposure to standard media channels—who are disproportionately impoverished, rural, and ethnically minoritized—are effectively excluded from the primary circuits of health communication. Ethnic minority women report the lowest awareness (36.1%) and near-zero vaccination (1.6%), reflecting the deep marginalization of Vietnam's remote populations [6]. This divide necessitates community-based outreach leveraging local health workers and peer networks [28].

4.1 Strengths and limitations

This study draws on a large (N = 4,102), nationally representative survey with a well-characterized sampling design, enabling robust population-level inference for young Vietnamese women. The use of survey-weighted Poisson regression yields directly interpretable prevalence ratios, circumventing the overestimation of risk inherent in odds ratios for prevalent outcomes like awareness [30]. The Erreygers decomposition framework provides precise, policy-actionable quantification of the specific drivers of inequality.

However, several limitations must be acknowledged. First, the cross-sectional design precludes causal inference. Second, outcome measures are self-reported and may be susceptible to recall or social desirability bias. Third, the MICS6 dataset lacks granular clinical details such as the number of doses received, timing relative to sexual debut, or reasons for non-vaccination. Fourth, by restricting the sample to women aged 15–29, the findings specifically characterize inequality among young women and may not generalize to older cohorts. Finally, ICT exposure was not excluded from models but its near-universal prevalence (97.9%) limits the ability to detect heterogeneity in digital information exposure. Future longitudinal research incorporating qualitative methodologies is required to decode the behavioral nuances of vaccine hesitancy and structural exclusion in Vietnam.

CONCLUSION AND RECOMMENDATIONS

Conclusion

Based on the analysis of 4,102 women aged 15–29 from the Vietnam MICS6 survey, this thesis demonstrates that the current paradigm of HPV vaccination in Vietnam is fundamentally inequitable.

1. The Knowing–Doing Chasm: While 62.4% of young women are aware of the HPV vaccine, only 7.5% have received it. This 55-percentage-point gap indicates that information diffusion, without concurrent financial accessibility, remains an insufficient public health strategy.

2. Severe Pro-Rich Inequality: The concentration index for vaccination (CI = 0.389) drastically eclipses that of awareness (CI = 0.149). Wealth does not merely influence vaccine uptake; it dictates it.

3. The Architecture of Disparity: Erreygers decomposition isolates out-of-pocket cost as the ultimate barrier. Household wealth single-handedly drives 51.4% of the inequality in vaccine uptake, compounded by mass media exclusion (23.8%), rural isolation (17.1%), and educational deficits (12.3%).

Recommendations

To eliminate cervical cancer, Vietnam must transition from opportunistic private vaccination to universal public provision:

1. **Universal Public Financing.** The HPV vaccine must be immediately integrated into the national Expanded Programme on Immunization (EPI) with full state subsidies. Relying on private markets guarantees the systematic exclusion of the poorest quintiles.

2. **Dismantling Geographic and Ethnic Barriers.** Conventional media cam-

paigns fail remote populations. The Ministry of Health must deploy mobile vaccination units and village health workers to directly serve ethnic minority and rural communities, bypassing traditional health infrastructure bottlenecks.

3. **School-Based Delivery Systems.** To bypass adult socioeconomic barriers entirely, vaccination must be universally administered through the public school system, guaranteeing equitable access prior to sexual debut.

4. **Data-Driven Accountability.** Future longitudinal and provincial-level research is required to track whether subsequent policy interventions actually reduce the socioeconomic disparities mapped in this study, holding the health system accountable for equity.

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APPENDIX

Appendix A1. Sensitivity analysis: Logistic regression results

As a sensitivity analysis, logistic regression models were fitted to compare odds ratios (OR) with the prevalence ratios (PR) reported in the main analysis. Results are presented in Table 4.1.

Table 4.1. Sensitivity analysis: Adjusted odds ratios (OR) from logistic regression (N = 3,899)

	Awareness		Vaccination	
Variable	**OR**	**95% CI**	**OR**	**95% CI**
Rural (ref: Urban)				
Yes	0.776	0.589–1.022	0.754	0.505–1.127
Age group (ref: 25–29)				
15–19	0.230	0.160–0.332	0.467	0.259–0.841
20–24	0.715	0.550–0.930	1.178	0.771–1.801
Education (ref: Upper sec+)				
Primary or less	0.249	0.144–0.430	0.314	0.056–1.760
Lower secondary	0.588	0.448–0.771	0.403	0.211–0.771
Wealth quintile (ref: Q5)				
Q1 (poorest)	0.198	0.123–0.317	0.104	0.033–0.326
Q2	0.361	0.245–0.533	0.534	0.313–0.911
Q3	0.372	0.255–0.544	0.387	0.232–0.646
Q4	0.525	0.365–0.757	0.581	0.362–0.933
Mass media (ref: No)				
Yes	1.395	0.992–1.962	2.050	0.763–5.508
Ethnic minority (ref: No)				
Yes	0.709	0.521–0.965	0.729	0.284–1.870
Decision autonomy (ref: Low)				
High	1.737	1.219–2.474	1.691	0.860–3.325

OR = odds ratio. The direction and statistical significance of all associations are consistent with the Poisson regression results (Table 3.5), confirming the robustness of findings.

Appendix A2. Awareness–vaccination gap model

A Poisson regression model was fitted among aware women only (N = 2,133 with subpopulation restriction) to identify factors associated with the awareness–vaccination gap.

Appendix A3. MICS6 Questionnaire items used

Table 4.2. MICS6 questionnaire items used for outcome variable construction

Item	Question	Response options
CCP5	Have you ever heard of HPV vaccination?	1 = Yes, 2 = No
CCP6	Have you ever received the HPV vaccine?	1 = Yes, 2 = No, 8 = Don't know
SB1	How old were you when you first had sexual intercourse?	Age in years, 0 = Never
DV1A–E	Who usually makes decisions about [specific topics]?	1 = Respondent alone, 2 = Jointly/other

Appendix A4. Stata analysis code

The complete Stata do-file used for all analyses is available in the study repository at:

`HPV thesis/Code/hpv_analysis_main.do`

The code implements the full analytical pipeline including data cleaning, survey design specification, descriptive statistics, bivariate analysis, concentration index computation, multivariable Poisson and logistic regression, and Erreygers decomposition. All analyses were performed using Stata version 17 with survey-weighted estimation.